23MIS0263
SHRISTI GUPTA

Agile Development Process and Devops Lab

LAB FAT

SET 3

LIBRARY BOOK MANAGEMENT SYSTEM

Step1: Creating the App structure and adding the codes:

App.java

```
import java.util.*;

public class App {

    private Map<Long, String> books = new HashMap<>();
    private long nextId = 1;

    // ADDING NEW BOOK
    public Long addBook(String bookName) {
        if (bookName == null || bookName.trim().isEmpty()) {
            throw new IllegalArgumentException("Book name cannot be empty");
        }
        Long id = nextId++;
        books.put(id, bookName);
        return id;
    }

    // LENDING BOOK
    public String getBook(Long id) {
```

```
        return books.get(id);
    }

    // UPDATE BOOK
    public boolean updateBook(Long id, String newName) {
        if (books.containsKey(id)) {
            books.put(id, newName);
            return true;
        }
        return false;
    }

    // DELETE BOOK
    public boolean deleteBook(Long id) {
        return books.remove(id) != null;
    }

    public static void main(String[] args) {
        System.out.println("Library Book Management System running...");
    }
}
```

AppTest.java

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

public class AppTest {
```

```
private App app;

@BeforeEach
void setUp() {
    app = new App();
}

@Test
void testAddAndGetBook() {
    Long id = app.addBook("Java Programming");
    assertNotNull(id);
    assertEquals("Java Programming", app.getBook(id));
}

@Test
void testUpdateBook() {
    Long id = app.addBook("Old Book");
    boolean updated = app.updateBook(id, "New Book");

    assertTrue(updated);
    assertEquals("New Book", app.getBook(id));
}

@Test
void testDeleteBook() {
    Long id = app.addBook("Delete Me");
    boolean deleted = app.deleteBook(id);

    assertTrue(deleted);
    assertNull(app.getBook(id));
```

```
  }
}
```

Deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: library-app-deployment

spec:
  replicas: 2

  selector:
    matchLabels:
      app: library-app

  template:
    metadata:
      labels:
        app: library-app

    spec:
      containers:
        - name: library-app
          image: shristig17/library-app:latest
          imagePullPolicy: Always

          ports:
            - containerPort: 8080
```

```yaml
---
apiVersion: v1
kind: Service
metadata:
  name: library-app-service

spec:
  type: NodePort

  selector:
    app: library-app

  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
```

Dockerfile

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/library-app-1.0.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
```

```
    http://www.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.demo</groupId>
<artifactId>library-app</artifactId>
<version>1.0</version>
<packaging>jar</packaging>

<name>library-app</name>

<properties>
   <maven.compiler.source>17</maven.compiler.source>
   <maven.compiler.target>17</maven.compiler.target>
   <junit.version>5.10.1</junit.version>
</properties>

<dependencies>
   <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>${junit.version}</version>
      <scope>test</scope>
   </dependency>
</dependencies>

<build>
   <plugins>

      <plugin>
```

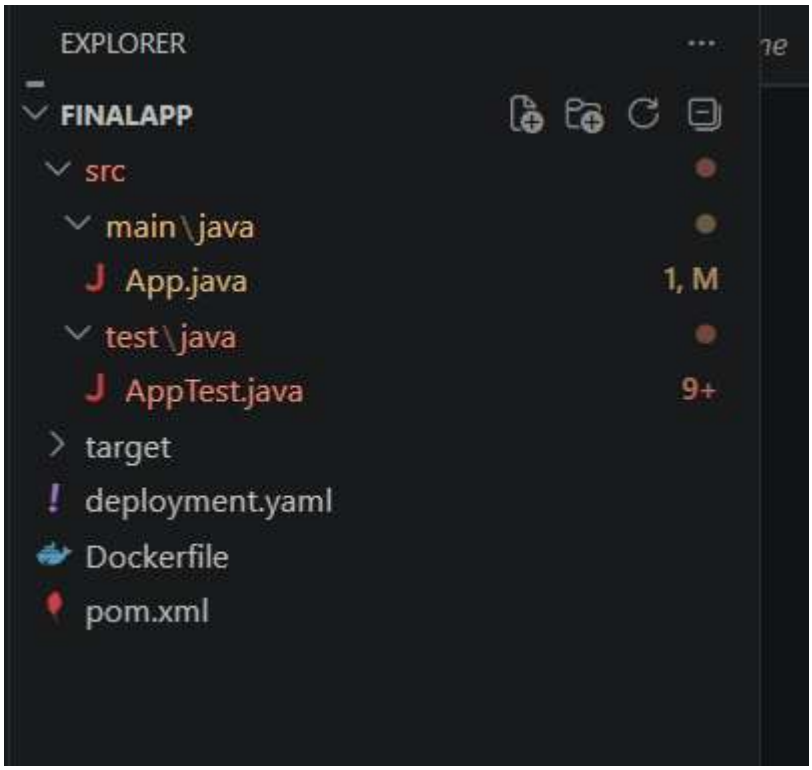
```
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>
        </plugin>

        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
            <configuration>
                <useModulePath>false</useModulePath>
            </configuration>
        </plugin>

        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.4.1</version>
            <configuration>
                <archive>
                    <manifest>
                        <mainClass>App</mainClass>
                    </manifest>
                </archive>
            </configuration>
        </plugin>

    </plugins>
</build>
```

</project>



Step 2: MVN tests to create target

```
PS C:\Users\vis_g\Downloads\finalapp> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< com.demo:library-app >------------------------
[INFO] Building library-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ library-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ library-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vis_g\Downloads\finalapp\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ library-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ library-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vis_g\Downloads\finalapp\src\test\resources
```

```
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ library-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.100 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.748 s
[INFO] Finished at: 2026-04-13T09:58:38+05:30
[INFO] ------------------------------------------------------------------------
```

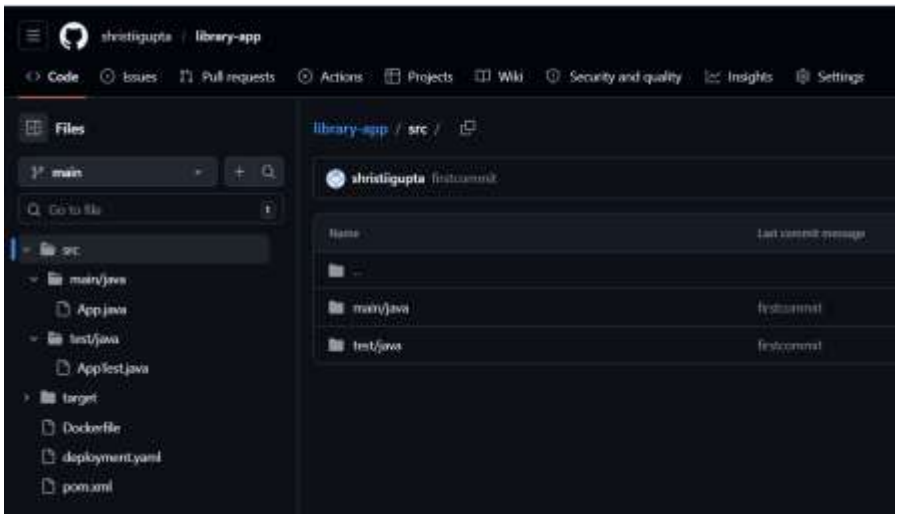
```
PS C:\Users\vis_g\Downloads\finalapp> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:library-app >-----------------------
[INFO] Building library-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ library-app ---
[INFO] Deleting C:\Users\vis_g\Downloads\finalapp\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ library-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vis_g\Downloads\finalapp\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ library-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform de
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
```

```
[INFO] --- resources:3.4.0:testResources (default-testResources) @ library-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vis_g\Downloads\finalapp\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ library-app ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ library-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.343 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ library-app ---
[INFO] Building jar: C:\Users\vis_g\Downloads\finalapp\target\library-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```

Step3: Pushing the code files to git after creating a repository on GitHub

Commands-

```
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/shristiigupta/library-app.git
git push -u origin main
```



Step 4: Creating pipeline

```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "shristig17/library-app:latest"
    }

    stages {
```

```
stage('Clone Repository') {
    steps {
        git branch: 'main',
            url: 'https://github.com/shristiigupta/library-app.git'
    }
}

stage('Build & Run Tests') {
    steps {
        bat 'mvn clean test'
    }
    post {
        always {
            junit 'target/surefire-reports/*.xml'
        }
    }
}

stage('Package Application') {
    steps {
        bat 'mvn package -DskipTests'
    }
}

stage('Build Docker Image') {
    steps {
        bat "docker build -t %DOCKER_IMAGE% ."
    }
}

stage('Push to DockerHub') {
    steps {
```

```
            withCredentials([usernamePassword(
                credentialsId: 'dockerhub',
                usernameVariable: 'DOCKER_USER',
                passwordVariable: 'DOCKER_PASS'
            )]) {
                bat """
                docker login -u %DOCKER_USER% -p %DOCKER_PASS%
                docker push %DOCKER_IMAGE%
                """
            }
        }
    }

    stage('Deploy to Kubernetes') {
        steps {
            bat 'kubectl apply -f deployment.yaml'
        }
    }
  }
}
```

General

Description

Library Book Management System - That lists and lends books along with some more functionalities

Plain text Preview

Step 5: Build Pipeline

```
[INFO] --- surefire:3.2.5:test (default-test) @ library-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.149 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.157 s
[INFO] Finished at: 2026-04-13T10:16:50+05:30
[INFO] ------------------------------------------------------------------------
Post stage
[Pipeline] junit
```

```
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ library-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ library-app ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ library-app ---
[INFO] Building jar: C:\Users\vis_g\.jenkins\workspace\library-mgmt\target\library-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.259 s
[INFO] Finished at: 2026-04-13T10:17:00+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat
```

```
latest: digest: sha256:5731350db8f5ea27acd39b3928a4a20807cb2d649a354404f018c30bacc40059 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\Users\vis_g\.jenkins\workspace\library-mgmt>kubectl apply -f deployment.yaml
deployment.apps/library-app-deployment created
service/library-app-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Tests 3

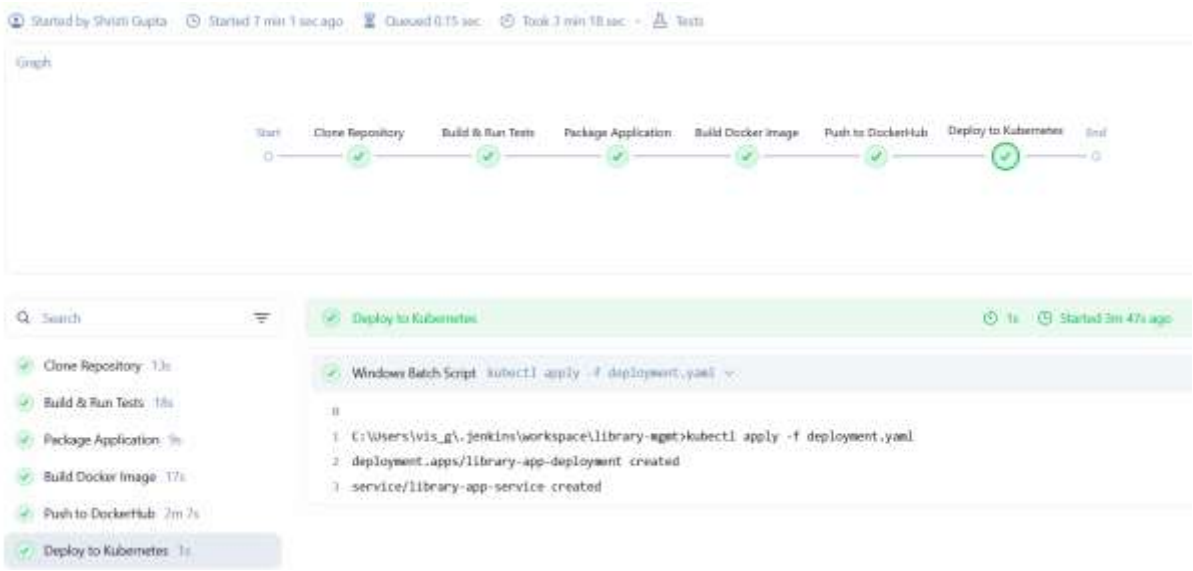
✓ 3 Took 0.14 sec

All tests are passing!

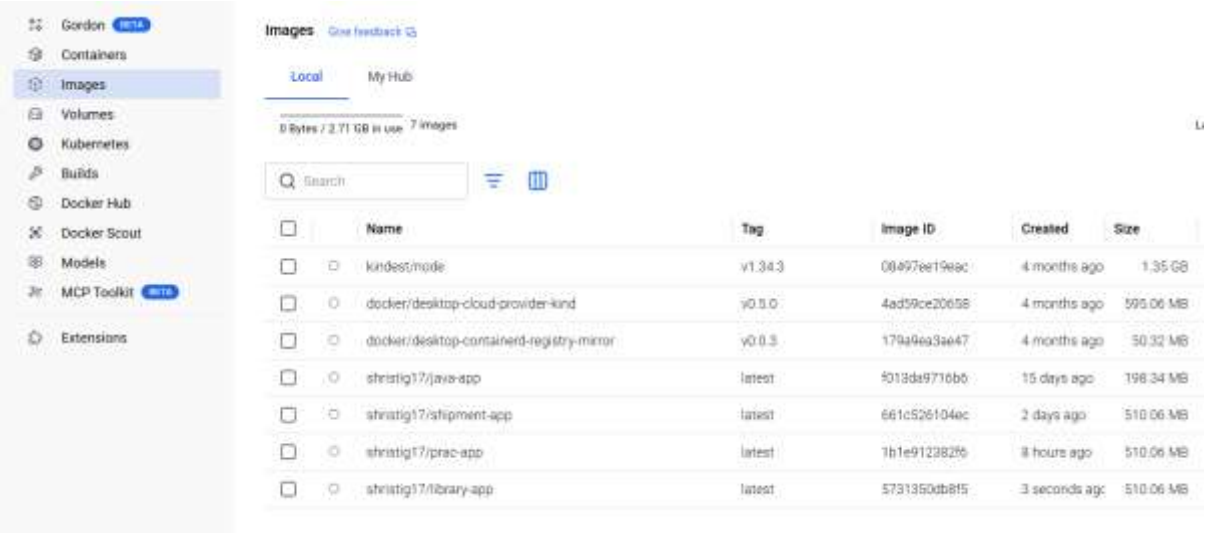
Nice one! All 3 tests are passing.

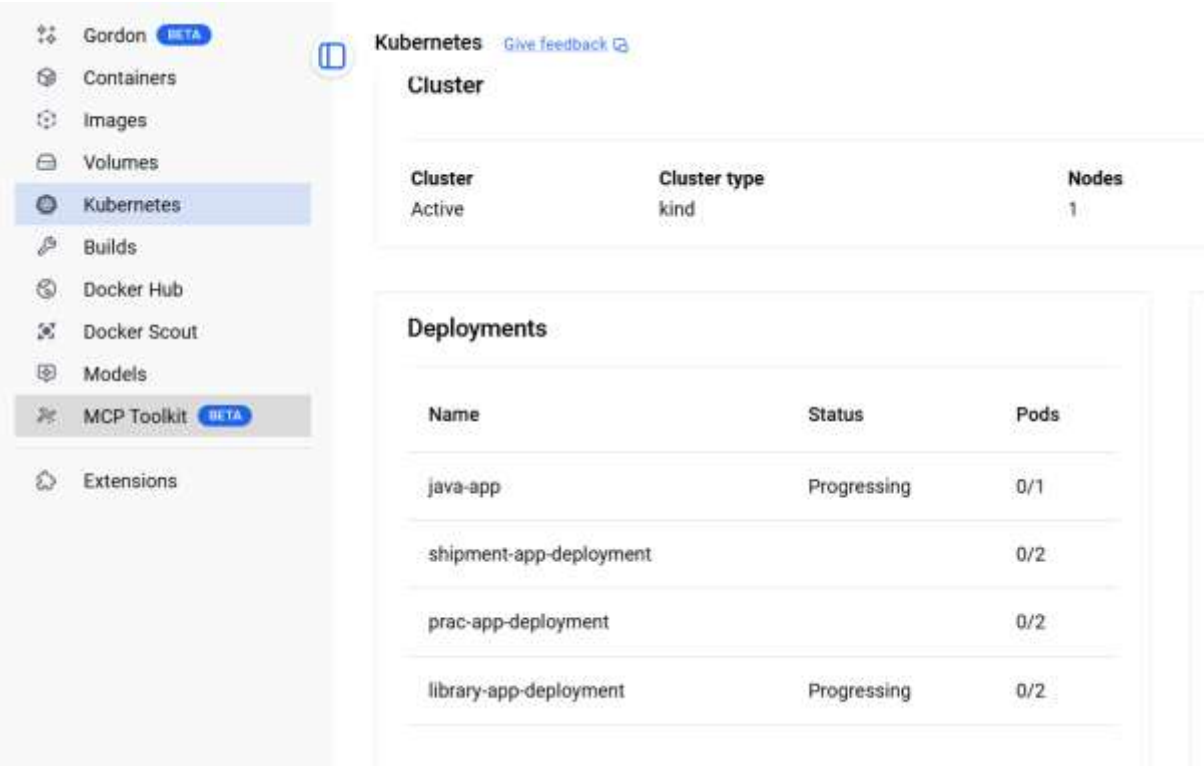
All Tests

Package	Failed	Skipped	Passed	Total	Duration
(root)	0	0	3 +3	3 +3	65 ms



Step 6: Checking on docker and Kubernetes:





Testing using - kubectl get svc

```
PS C:\Users\vis_g\Downloads\finalapp> kubectl get svc
NAME                  TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        14d
library-app-service   NodePort    10.96.59.241    <none>        80:30285/TCP   13m
prac-app-service      NodePort    10.96.136.197   <none>        80:31560/TCP   8h
shipment-app-service  NodePort    10.96.63.203    <none>        80:30007/TCP   2d12h
PS C:\Users\vis_g\Downloads\finalapp>
```